Disease Detection System

Mock Defense Script — Student & Panel Dialogue · Capstone Project Defense Practice

PANEL

Good afternoon. Please tell us what problem your project solves.

STUDENT

Good afternoon, and thank you. When someone dies, the information about that death exists immediately, but it does not become intelligence immediately. By the time district and national epidemiologists can see a pattern, an outbreak has already had days or weeks to spread. The gap is not in recording deaths. It is in the latency between a death occurring and that death becoming actionable information for the people who respond to it. My system closes that gap.

PANEL

So what did you build?

STUDENT

The Disease Detection System. A real-time mortality surveillance platform. Death notifications flow in from facility officers on web and mobile, including offline field capture that syncs later, and the system continuously aggregates, forecasts, and scans them for anomalies. It answers three questions for public health decision-makers: how many deaths are coming, is something abnormal happening right now, and what is killing people.

PANEL

Those three questions sound like system objectives. State them clearly.

STUDENT

Three objectives. One, predict mortality trends: the system forecasts expected deaths per period, overall, per cause category, and per district, with confidence ranges. Two, detect outbreaks early: threshold clusters, low-volume early warnings using a Poisson CUSUM, instant MPDSR alerts for maternal and perinatal deaths, and a community-signal layer that

can alert before a death is even notified. Three, identify cause-of-death patterns: ICD-10 coding from the preliminary report through to the autopsy- or certification-verified final cause.

PANEL

Walk us through how it works.

STUDENT

I sign in as a public health analyst. On the dashboard I see live key indicators: deaths in the last seven days, pending reviews, and active alerts. A facility officer enters a death notification, and it is ICD-10 coded at capture. Within minutes, the outbreak engine scans every notification in a rolling window. Threshold clusters and early warnings appear on the alerts page and push to the notification bell in real time. On the predictions page, the system shows forecasts per category and per district with confidence ranges, using a seasonal model when there is enough history. And when a pathologist finalizes an autopsy, the verified cause replaces the preliminary cause everywhere, including detection and prediction.

PANEL

A mortality system only sees an outbreak after people have died, surely?

STUDENT

Not entirely — and closing that gap was a deliberate design decision. Mortality notifications are the backbone, but the engine has two pre-mortality passes. First, a community-signal pass: reports from health workers, police, media and the public over USSD and WhatsApp aggregate per district — three reports, or one critical one, raise an early warning. Second, a syndromic pass: facilities report aggregate case counts — diarrhoea, respiratory illness, rash fever — over REST or FHIR, and ten pooled cases of a syndrome in a district in a week raise an alert with zero deaths on file. And late data entry cannot hide a cluster: the engine counts a death from when the system learned of it, not only when it occurred.

PANEL

Why not simply use DHIS2, which Zimbabwe already runs?

STUDENT

DHIS2 aggregates. It does not do per-case autopsy verification, real-time anomaly detection, or offline field capture. My system is designed to sit inside that ecosystem, not beside it: it pulls from and pushes to DHIS2 through the configured data set, and it can route through OpenHIM middleware. It is a partner to DHIS2, not a competitor.

PANEL

How do you protect the data of the deceased?

STUDENT

Role-based access control with facility scoping, Argon2id password hashing, stateful server-side sessions that expire after thirty minutes idle, two-factor authentication, a full audit trail of every action, and encrypted offline storage on the mobile app.

PANEL

What happens when your machine learning service fails?

STUDENT

Every analytics endpoint has a Node fallback. Detection and prediction degrade gracefully, they never vanish. The design assumption was that in the field, partial function is far better than none.

PANEL

What are the honest limitations?

STUDENT

Two. First, the forecast confidence bands are approximate residual intervals, not validated probabilistic bounds. Second, real deployment needs production email and DHIS2 credentials, which are configured but currently point at test accounts. I would rather name those limitations myself than pretend they do not exist.

PANEL

Thank you. We will deliberate.

STUDENT

Thank you. I welcome your questions.

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